

Hoja de Trabajo 1

$$1) \int_0^3 (1 + |x-1|) dx = \int_0^1 1 - x + 1 + \int_1^3 1 + x - 1$$

$$\begin{aligned} x-1 &= 0 \\ x &= 1 \end{aligned}$$

$$\int_0^1 (2-x) dx + \int_1^3 (x) dx = 2x - \frac{x^2}{2} \Big|_0^1 + \frac{x^2}{2} \Big|_1^3$$

$$\begin{aligned} (0,1) & (1,3) \\ 0.5 & 1.5 \\ & + \end{aligned}$$

$$[2(1) - \frac{1^2}{2}] + [\frac{3^2}{2} - \frac{1^2}{2}] = \frac{3}{2} + 4 = \underline{\underline{5.5}}$$

$$2) V(t) = t^2 - t - 12 \quad (\text{m/s}^2) \quad (7.5)$$

$$V(t) = 0$$

$$t^2 - t - 12 = 0$$

$$(t+3)(t-4) = 0$$

$$t = -3 \quad t = 4$$

$$\begin{aligned} (1,4) & (4,5) \\ 1.5 & 4.5 \\ & + \end{aligned}$$

$$-\int_1^4 (t^2 - t - 12) dt + \int_4^5 (t^2 - t - 12) dt$$

$$-\left(\frac{t^3}{3} - \frac{t^2}{2} - 12t\right) \Big|_1^4 + \left(\frac{t^3}{3} - \frac{t^2}{2} - 12t\right) \Big|_4^5$$

$$-\left(\frac{4^3}{3} - \frac{4^2}{2} - 12(4)\right) + \left(\frac{5^3}{3} - \frac{5^2}{2} - 12(5)\right) - \left(\frac{1^3}{3} - \frac{1^2}{2} - 12(1)\right)$$

$$\left[\left(-\frac{4^3}{3} + \frac{4^2}{2} + 12(4)\right) - \left(-\frac{1^3}{3} + \frac{1^2}{2} + 12(1)\right)\right] + \left[\left(\frac{5^3}{3} - \frac{5^2}{2} - 12(5)\right) - \left(\frac{4^3}{3} - \frac{4^2}{2} - 12(4)\right)\right]$$

$$\frac{704}{3} - \frac{73}{6} + \left(-\frac{785}{6}\right) - \left(-\frac{104}{3}\right)$$

$$\frac{45}{2} + \frac{23}{6} = \frac{79}{3} \approx \underline{\underline{26.33}}$$

$$3) \int_0^{x^2} \sin^2 \theta d\theta = \frac{d}{dx} \int_0^{x^2} \sin^2 \theta d\theta = \left(\frac{d}{dx} \int_0^u \sin^2 \theta d\theta\right) = 2x \cdot \frac{d}{du} \int_0^u \sin^2 \theta d\theta$$

$$u = x^2$$

$$= 2x \sin u^2 = 2x \sin x^4$$

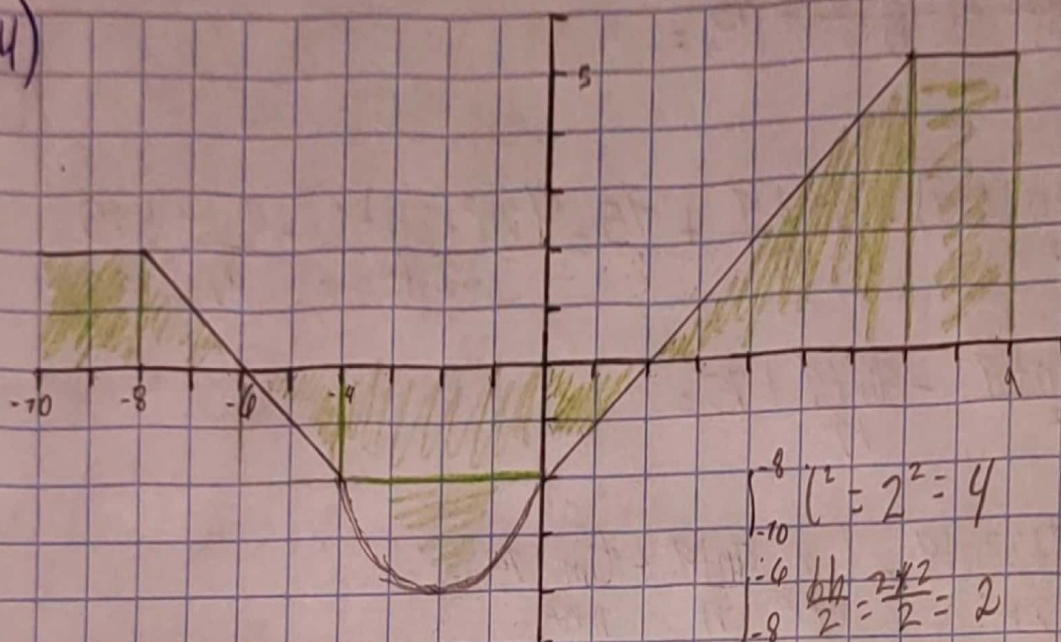
$$\frac{du}{dx} = 2x$$

$$x = \frac{\pi}{3}$$

$$2\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{3}\right)^4 = \underline{\underline{7.954}}$$

$$\frac{du}{2x} = dx$$

4)



$$\int_{-10}^{-8} 4 \, dx = 4 \cdot 2 = 8$$

$$\int_{-8}^{-6} \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_{-6}^{-4} \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_{-4}^{-2} \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_{-2}^0 \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_0^2 \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_2^4 \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_4^9 5 \, dx = 5 \cdot 5 = 25$$

$$\int_{-10}^9 f(x) \, dx = 8 + 2 + 2 + 2 + 2 + 2 + 25 = 43$$

$$\int_{-10}^9 f(x) \, dx = 43$$

$$\int_{-4}^0 bh = 4 \cdot 2 = 8 + \int_{-4}^0 \frac{\pi r^2}{2} = -2\pi = -14.28$$

$$\int_0^2 \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_2^4 \frac{bh}{2} = \frac{2 \cdot 2}{2} = 2$$

$$\int_4^9 bh = 2 \cdot 5 = 10$$

$$5) \int_{-2}^1 (x^2 + 2x + 5) \, dx$$

$$\Delta x = \frac{1 - (-2)}{n} = \frac{3}{n} \quad x_i = -2 + \frac{3i}{n}$$

$$f(x_i) = \left(-2 + \frac{3i}{n}\right)^2 + 2\left(-2 + \frac{3i}{n}\right) + 5 = \left(\frac{3i}{n} - 2\right)^2 + 2\left(\frac{3i}{n} - 2\right) + 5 =$$

$$= \frac{9i^2}{n^2} - \frac{12i}{n} + 4 + \frac{6i}{n} - 4 + 5 = \frac{9i^2}{n^2} - \frac{6i}{n} + 5$$

$$\int_{-2}^1 (x^2 + 2x + 5) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{9i^2}{n^2} - \frac{6i}{n} + 5 \right) \left(\frac{3}{n} \right) = \frac{27}{n^3} - \frac{18}{n^2} + \frac{15}{n}$$

$$= \lim_{n \rightarrow \infty} \frac{27}{n^3} \sum_{i=1}^n i^2 - \frac{18}{n^2} \sum_{i=1}^n i + \frac{15}{n} \sum_{i=1}^n 1 = \lim_{n \rightarrow \infty} \frac{27}{n^3} \left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{18}{n^2} \left(\frac{n(n+1)}{2} \right) + \frac{15}{n} (n)$$